SEOK JOON KIM

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EDUCATION

MSc Robotics, Systems and Control, ETH Zürich, Switzerland (GPA: 5.63/6) Sep. 2019 - Feb. 2022

Relevant modules: Autonomous Mobile Robots, Robot Dynamics, Vision Algorithms for Mobile Robots, Recursive Estimation, Dynamic Programming and Optimal Control, Model Predictive Control, VR I/II, Intro. to ML

BSc Aerospace Engineering, TUDelft, the Netherlands (GPA: 8.51/10, CUM LAUDE) Sep. 2016 - Aug. 2019

Selected for the Honours Program (Top 5% of first-year students).
Minor in Electronics for Robotics

WORK EXPERIENCE

Neuromeka Co. Ltd., Seoul, South Korea October 2024 - Ongoing

Full-Time, Alternative Military Service

Predictive Human-Robot Collaboration with Mobile Manipulators

- Developing predictive control for manipulators to proactively assist humans in step-by-step tasks by analyzing Hand-Object Interaction, anticipating intentions, and ensuring safe, efficient collaboration.
- MPPI-based navigation with local mapping and dynamic obstacle avoidance.

MAXST Co. Ltd., Seoul, South Korea Mar. 2022 - Sep. 2024

Full-Time, Alternative Military Service

3D Scene Understanding & CAD Model Retrieval (Team Leader)

- Researched 3D scene understanding for object detection and segmentation in indoor spaces.
- Developed CAD model retrieval to replace detected objects with accurate 3D models.
- Led a team of three in 3D computer vision research and development.

3D Reconstruction of Indoor Spaces (Project Manager)

- Led the development of MAXSCAN, an iOS app for mobile 3D room reconstruction for VR/AR.
- Developed an interactive, user-guided approach for room layout reconstruction using RGB-D data.
- Researched mobile-friendly texturing and image inpainting to enhance 3D realism.
- Managed a team of five (designer, researchers, developers) from proof-of-concept to productization.

AR Scanner

- Developed image retrieval and relocalization modules for large-scale AR experiences.
- Implemented mesh splitting and texturing methods for efficient rendering on mobile devices.

The German Aerospace Center (DLR), Germany Dec. 2020 - May 2021

Internship

3D Point Cloud Registration for Ego-Motion Estimation

- Developed deep learning models for 3D point cloud registration to estimate drone ego-motion with a robotic arm using simulated LiDAR data.

Gaussian Process Uncertainty Estimation

- Researched deep neural network architectures for Visual-Inertial Odometry (VIO) and Visual Odometry (VO), trained on the KITTI dataset.
- Simulated sensor degradation (IMU noise, gyro biases, camera distortions) to analyze robustness.
- Integrated Gaussian Process (GP) experts within the DeepVIO to estimate uncertainty in pose predictions

RELEVANT PROJECTS

Personal Project, South Korea

Computer Vision Solution for Worker Safety in Manufacturing Plants Aug. 2024 - Present

- Developed a semi-automatic video labeling tool using vision grounding and continual learning to detect uncommon classes and enhance annotation efficiency.
- Utilizing MonoDepth and 2D segmentation models to reconstruct 3D bounding boxes and compute their intersection with machinery to assess worker proximity and potential hazards.

ETH Zürich, Switzerland

Master's Thesis: Conditional Hand Motion Synthesis Jul. 2021 - Jan. 2022
Advanced Interactive Technologies Lab, Prof. Otmar Hilliges

- Designed a CVAE-based neural network to generate realistic human hand motions for object interactions.
- Developed an efficient data preprocessing pipeline to segment motion frames into approach, grasp, and action phases based on contact and object movement.
- Proposed novel evaluation metrics to assess the static and dynamic realism of generated motions.

Semester Project: Viewpoint-Tolerant Place Recognition with Semantic Segmentation Sep. 2020 - Jan. 2021
Vision for Robotics Lab, Prof. Margarita Chli

- Improved place recognition under large viewpoint variations by integrating semantic segmentation and keypoint filtering, enhancing PR and AUC metrics.
- Developed an automatic binary labeling system to classify image pairs as "same place" or "different place" using ground truth depth and pose data, generating a benchmark dataset.
- Implemented evaluation tools to assess PR and AUC performance across different algorithms.

PUBLICATIONS AND PATENTS

Kim, S. J., et al. (Oct. 2024). RoomRecon: High-Quality Textured Room Layout Reconstruction on Mobile Devices. IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Seattle, WA.
DOI: 10.1109/ISMAR62088.2024.00069. Best Paper Nominee.

Duc, C. D., **Kim, S. J.**, & Cho, K. S. (Nov. 2024). Geometric Exploitation for Indoor Panoramic Semantic Segmentation. *NeurIPS*, Vancouver, Canada)

Lee, J. H., Kim, Y. H., **Kim, S. J.**, et al. (May. 2025). Learning Fast, Tool-aware Collision Avoidance for Collaborative Robots. *IEEE Robotics and Automation Letters*, Accepted

Choi, S. W., Chae, H. W., Jeung, Y. S., **Kim, S. J.**, Cho, K. S., & Kim, T. W. (2023). Fast and versatile feature-based LiDAR odometry. *IEEE Robotics and Automation Letters*, 8(2), 640-647.

Accepted Patent: Method, computer program, and electric device for generating 3D model based on plane information recognized according to user input. South Korean Patent No. 1026224310000 (Released: January 3, 2024).

SCHOLARSHIPS

Excellence Scholarship & Opportunity Programme (ESOP) Sep. 2019 - Aug. 2021

- Awarded the fully funded Excellence Scholarship by the ETH Zürich Foundation, granted to 51 students among all incoming Master's students, covering the full duration of the Master's program.

Swiss-European Mobility Programme (SEMP) Dec. 2020 - May. 2021

- Awarded for an internship at the German Aerospace Center (DLR) under Erasmus+ for higher education.

EXTRA-CURRICULAR ACTIVITIES

IDEA League Challenge Participant (TU Delft) Aug. 2018 - June 2019

- Selected as one of six students to represent TU Delft in the IDEA League Challenge, a collaborative program with top European universities.
- Engaged in multi-weekend challenges, working with diverse teams to address societal, technological, and political challenges, emphasizing multidisciplinary problem-solving beyond technical aspects.

First-Year Student Mentor and Teaching Assistant (TU Delft) Aug. 2018 - Feb. 2019

- Guided two student teams through first-year projects by explaining concepts, assisting with wind tunnel testing, and grading reports.
- Mentored 21 first-year students, supporting their academic transition and integration through regular individual meetings.

SKILLS

Computer Languages	Python, C++, Swift & Objective C++ (iOS)
Software & Tools	Pytorch, Tensorflow, Git, Bash, Docker
Spoken Languages	Korean (native), English (proficient), German (beginner)